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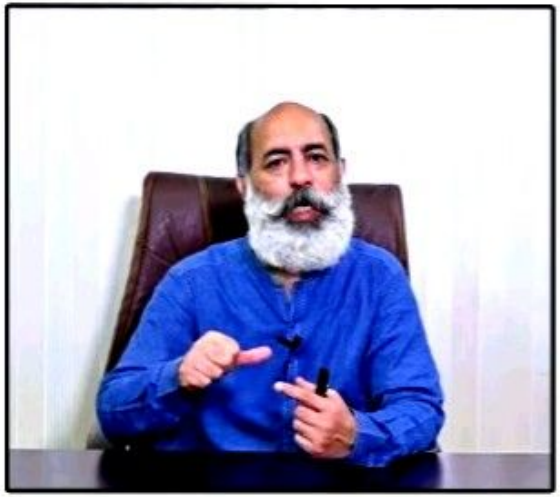
Faisalabad

Applications of Information and Communication Technologies



Dr. Muhammad Irfan Khan

lining made on the hands of human beings

Components of Data in ICT

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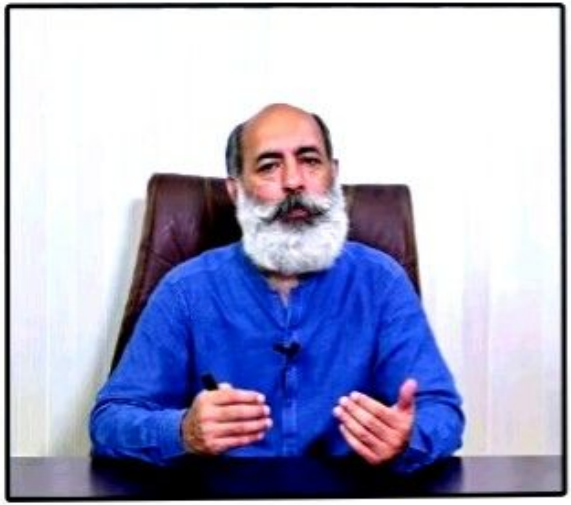
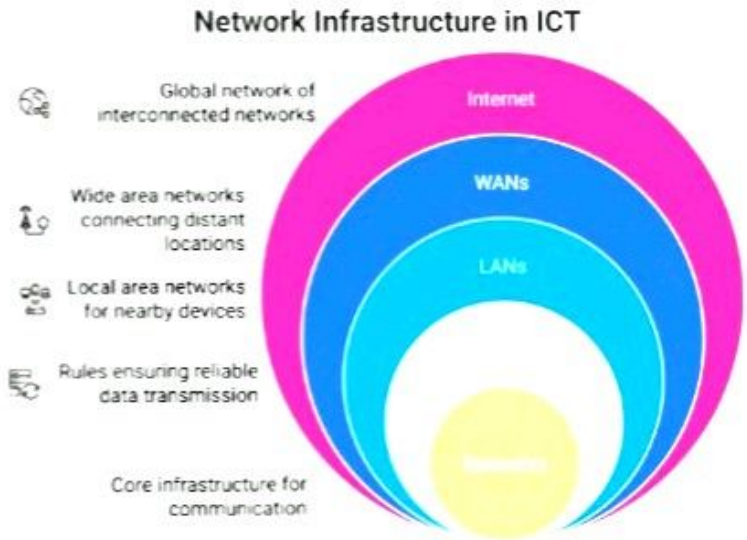
Applications of Information and Communication Technologies



Dr. Muhammad Irfan Khan



interacts with each other. The popular protocols among these protocols

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Applications of Information and Communication Technologies

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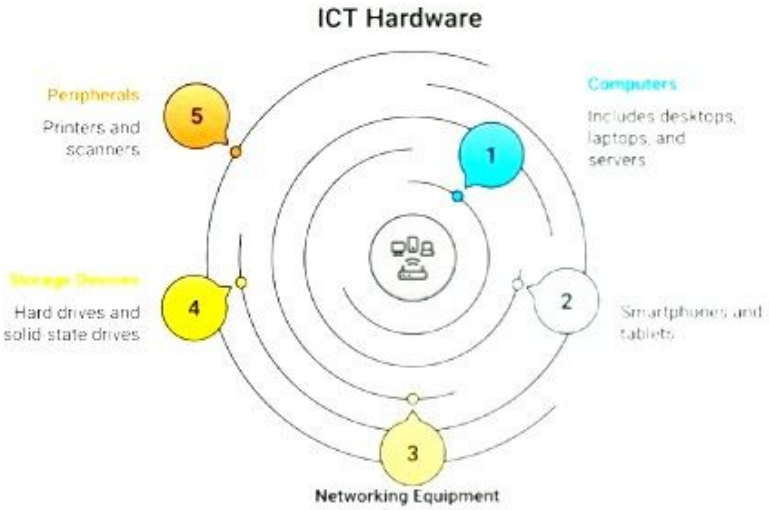
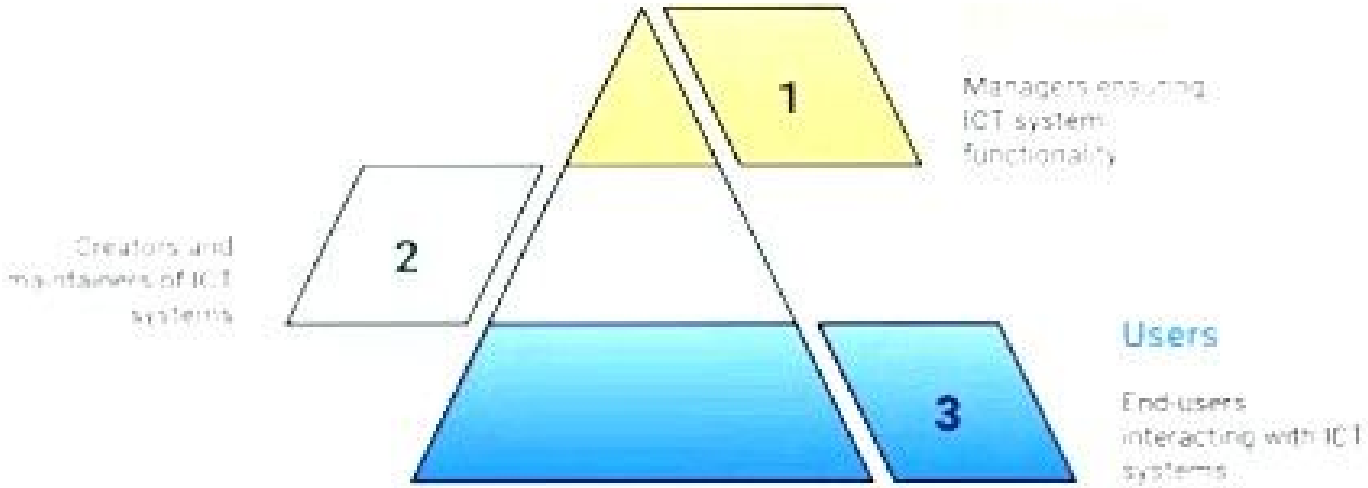
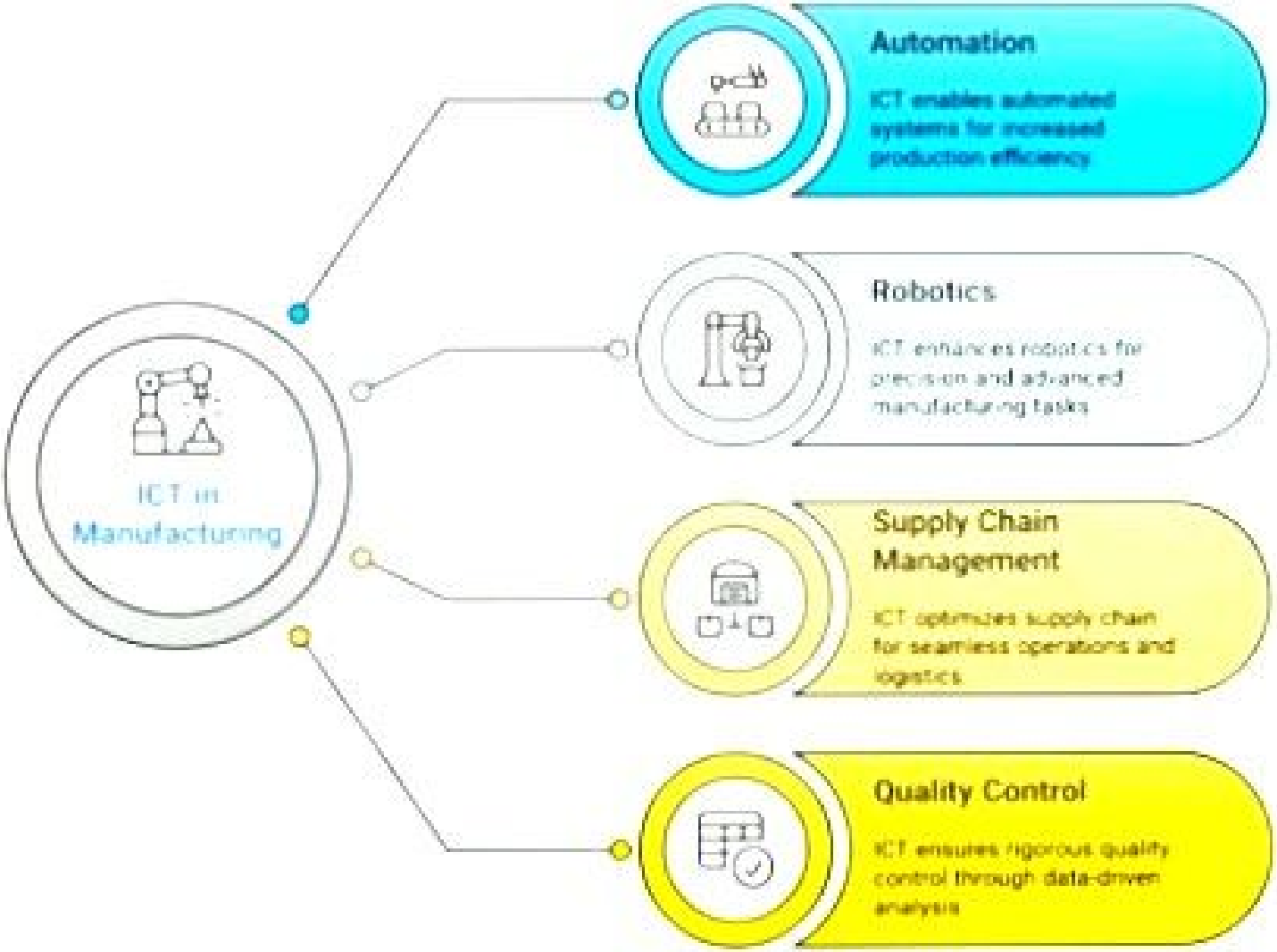
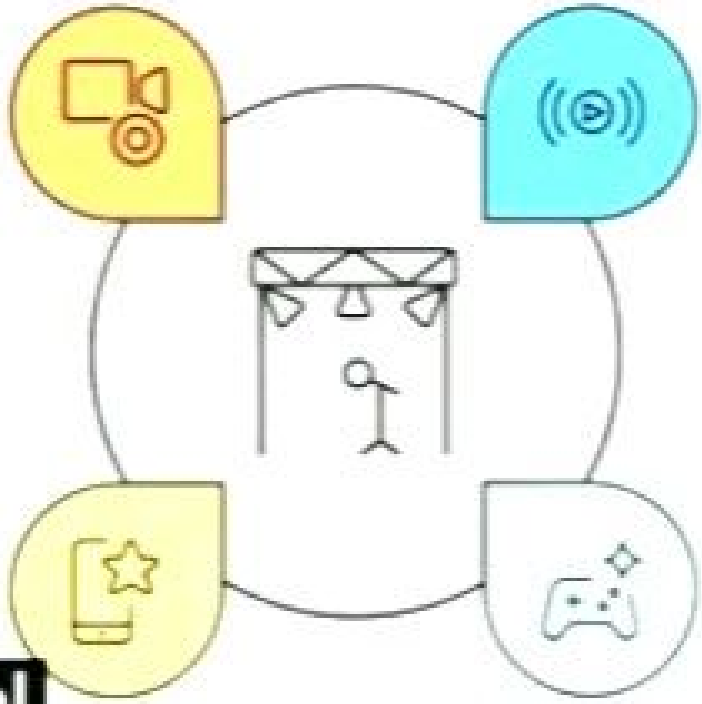


ICT People Hierarchy




ICT in Manufacturing
Automation
ICT enables automated systems for increased production efficiency
Robotics
ICT enhances robotics for precision and advanced manufacturing tasks
Supply Chain Management
ICT optimizes supply chain for seamless operations and logistics
Quality Control
ICT ensures rigorous quality control through data-driven analysis

ICT's Role in Entertainment



Digital Content Creation

ICT supports the production of digital media

Streaming Services

ICT enables on-demand video and music platforms

ICT facilitates global communication and

ICT enhances interactive gaming with real-time connectivity

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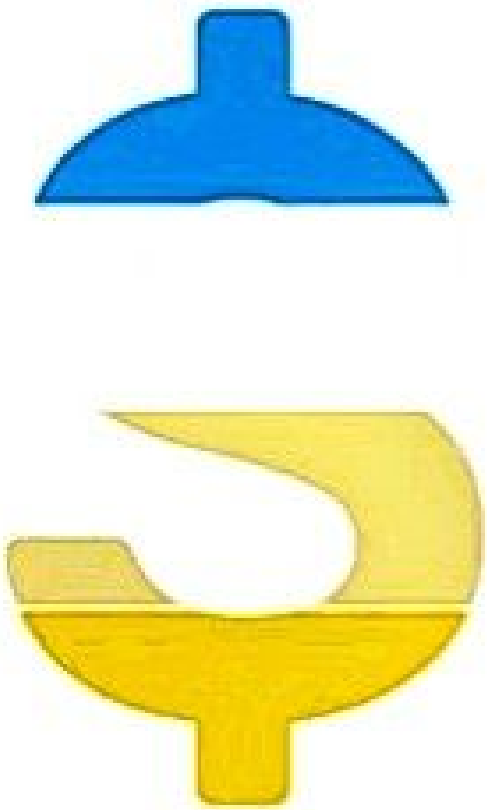
ICT's Role in Finance



Online Banking

Facilitates digital transactions and account management

Enables real-time market access and trading platforms

Fraud Detection

Enhances security measures to prevent financial crimes

Risk Management

Improves assessment and mitigation of financial risks

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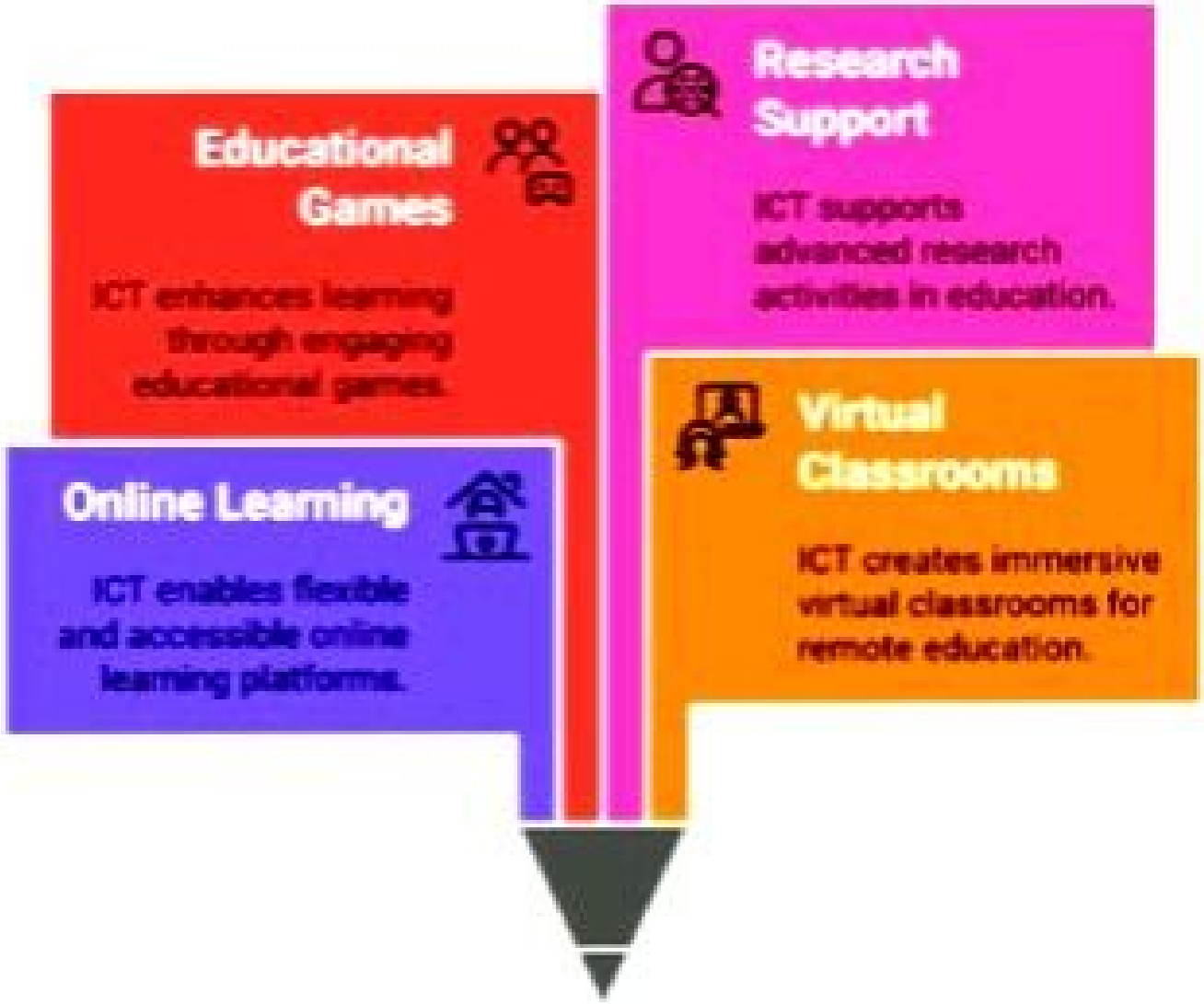
ICT's Impact on Business

ICT's Role in Education



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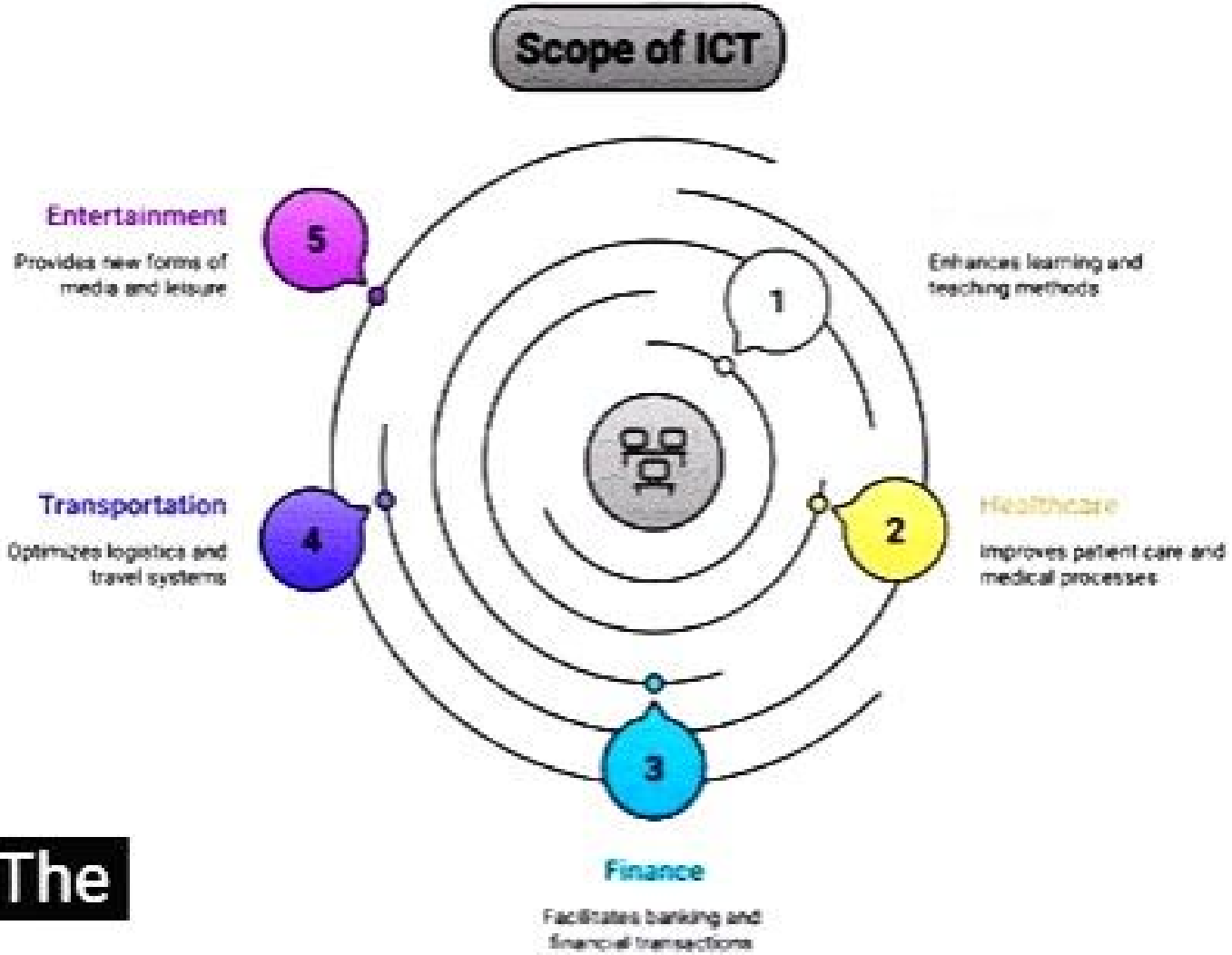
Scope of ICT
Entertainment
Provides new forms of media and leisure
5
1
Enhances learning and teaching methods
Transportation
Optimizes logistics and travel systems
4
2
Improves patient care and medical processes
3
Finance
Facilitates banking and financial transactions

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Data privacy and security

Pros

1

Implementing safeguards enhances protection of personal information from cyberattacks.

2

Building trust and confidence in data handling practices.

3

Ensuring compliance with data privacy and security regulations.

Cons

1

Cost of implementation

High costs associated with implementing advanced security measures.

2

Complexity of systems

Complex systems require specialized knowledge and management.

3

Potential for misuse

Risk of data misuse despite security measures.

Challenges and Opportunities of ICT Implementation

Pros

- Enhanced communication
- Increased productivity
- Economic growth
- Improved education
- er healthcare

Cons

- Digital divide
- Cybersecurity threats
- Rapid obsolescence
- Privacy concerns
- Social inequalities

here are

Conclusion

Information and Communication Technologies (ICT) are fundamental to modern society, impacting nearly every aspect of our lives.

Understanding the components, scope, and emerging technologies within ICT is crucial for individuals and organizations alike.

As technology continues to evolve, staying informed and adapting to new

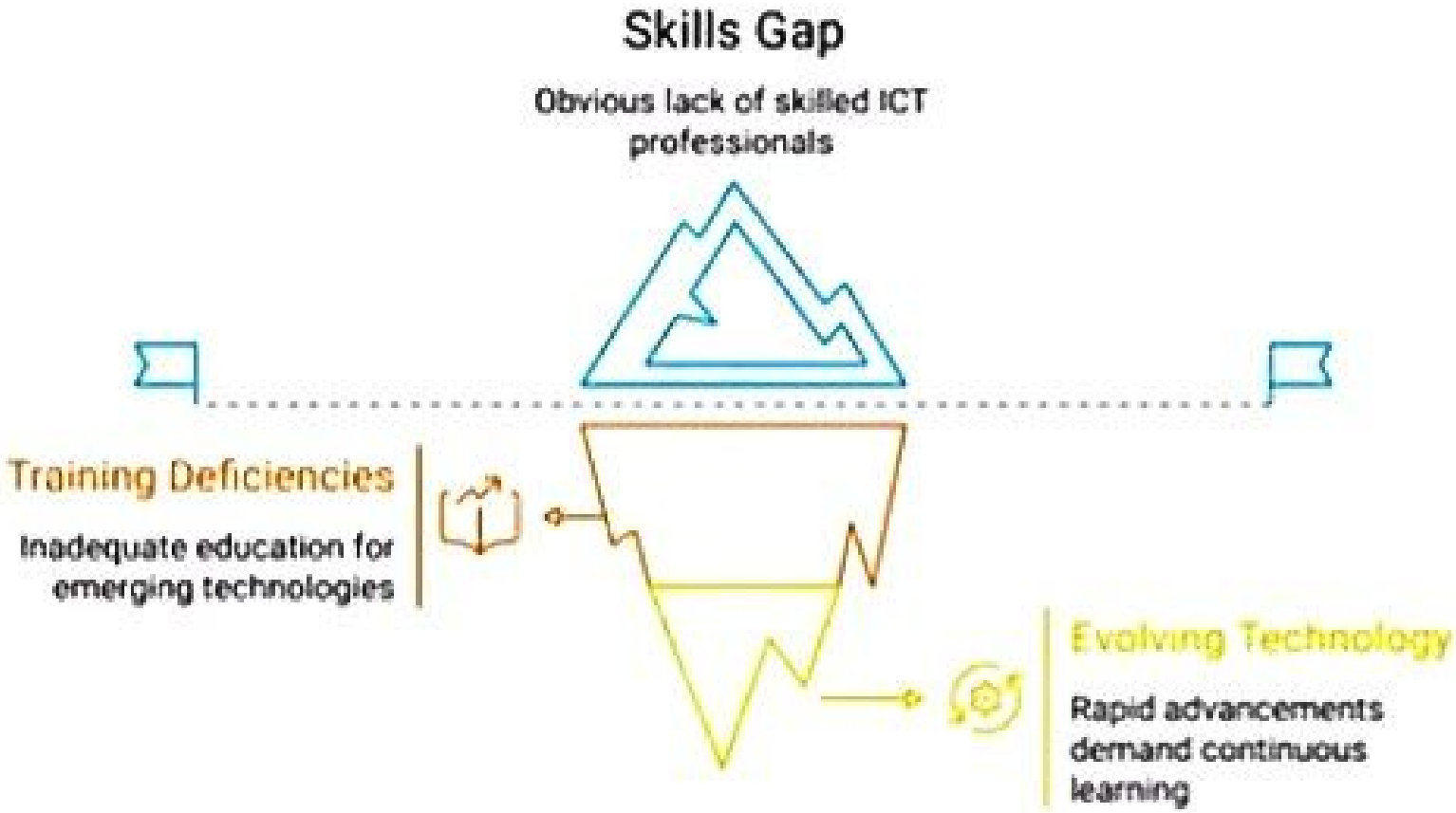
was lopments will be essential for success in the digital age.

Good news . . .

Despite these challenges, ICT offers great opportunities for innovation, economic growth, and social progress.

By addressing the challenges and leveraging the opportunities, we can apply the **power of ICT** to create a better future for all.

ICT Skills Gap: Unveiling the Underlying Issues



Ethical Considerations in ICT

Misinformation Spread

Misinformation spread is complex but has low impact due to its manageable nature

Bias in AI

Bias in AI has high impact and complexity due to its potential for widespread discrimination

Automation on Employment

Automation on employment is simple with low impact, affecting specific job sectors

Privacy Concerns

Privacy concerns are impactful but less complex, requiring straightforward solutions.

Questions

6. Explain the scope of ICT in modern life.
7. How is ICT used in the education sector?
8. Describe how ICT benefits the healthcare industry.
9. What role does ICT play in business or commerce?
10. Give examples of how ICT is used in government services.

ns,

1. What does the term ICT stand for?
2. Shortly describe the core components of ICT.
3. What are the main components of ICT hardware?
4. What is the function of network infrastructure in ICT?
5. What are the main components of data in ICT?

Questions

11. What are some common challenges in implementing ICT in developing countries?
12. Discuss the digital divide as a challenge in ICT deployment.